STACKS, DISCRIMINANTS: LEMMA-LOCALLY-COMES-FROM-UNIVERSAL-FINITE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2508-2534 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 2536-2551, the complete author proof (14 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 32 of queue_discriminant.json (pb=844, stmt 2508-2534, proof 2536-2551).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0FKZ: sources/stacks-discriminant/pinned-original-source.tex, source lines 2326–2339. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FKY: sources/stacks-discriminant/pinned-original-source.tex, source lines 2282–2298. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FL0: sources/stacks-discriminant/pinned-original-source.tex, source lines 2363–2368. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : Y \to X$ be a morphism of schemes. If f satisfies the equivalent conditions of Lemma 0FKY then for every $x \in X$ there exist a d and a commutative diagram*

$$\begin{array}{ccccccc} Y & \longleftarrow & V & \longrightarrow & V_d & \longrightarrow & Y_d \\ \downarrow & & \downarrow & & \downarrow & & \downarrow{\scriptstyle \pi_d} \\ X & \longleftarrow & U & \longrightarrow & U_d & \longrightarrow & X_d \end{array}$$

with the following properties

(1) *$U \subset X$ is open, $x \in U$, and $V = f^{-1}(U)$,*
(2) *$\pi_d : Y_d \to X_d$ is as in Example 0FKZ,*

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(3) $U_d \subset X_d$ *is as in Lemma 0FL0 and* $V_d = \pi_d^{-1}(U_d) \subset Y_d$,
(4) *where the middle square is cartesian.*

Proof. Choose an affine open neighbourhood $U = \mathrm{Spec}(A) \subset X$ of x. Write $V = f^{-1}(U) = \mathrm{Spec}(B)$. Then B is a finite locally free A-module and the inclusion $A \subset B$ is a locally direct summand. Thus after shrinking U we can choose a basis $1 = e_1, e_2, \ldots, e_d$ of B as an A-module. Write $e_i e_j = \sum \alpha_{ij}^l e_l$ for unique elements $\alpha_{ij}^l \in A$ which satisfy the relations $\sum_l \alpha_{ij}^l \alpha_{lk}^m = \sum_l \alpha_{il}^m \alpha_{jk}^l$ and $\alpha_{ij}^k = \alpha_{ji}^k$ and $\alpha_{i1}^j = \delta_{ij}$ in A. This determines a morphism $\mathrm{Spec}(A) \to X_d$ by sending $a_{ij}^l \in A_d$ to $\alpha_{ij}^l \in A$. By construction $V \cong \mathrm{Spec}(A) \times_{X_d} Y_d$. By the definition of U_d we see that $\mathrm{Spec}(A) \to X_d$ factors through U_d. This finishes the proof. $\square$